Oral Check Assist — Clinician guide

How on-device screening works, what patients see, and how to use the app in practice

Screening adjunct only — not a substitute for clinical examination or histopathology

This document explains how the app decides **risk level** and **recommendations**, what the AI is validated to do (and what it is not), and how clinicians can use it effectively—including in **controlled validation** settings (for example, studies where examiners who already see tobacco-using patients also run the app and outcomes are compared to clinical judgment). Technical model-training detail is omitted here; this guide is self-contained for clinical and operational use.

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1. Overview

The app runs **entirely on the device**: photos are analyzed locally; there is no cloud inference step in the default screening flow.

It uses **two AI checks** on each photo:

1. **Screening check (binary)**
   Produces **Screening Risk Score** = P(abnormal). This drives whether the session is **screen-positive** or **screen-negative** at the app's operating point (see Section 5).

2. **Detail check (5-class)**
   Predicts a **label**: Healthy, Benign, OPMD, OCA, or Unknown. Used for **subtype** (wording and urgency) and **label confidence**; it does **not** override the binary screening decision.

Key principle: The binary model decides screening; the 5-class model refines the message and aggregated label. With **multiple photos**, the app applies a **hybrid rule** on binary scores (`maxScore ≥ 0.65` or `hybridAvg ≥ 0.4735` → screen positive) and aggregates 5-class over photos that contribute to that decision (Section 5).

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2. Using the app: walkthrough (what you and the patient do)

Typical flow in the mobile app:

1. **Preparation screen** — The patient (or assistant) confirms simple checks: good lighting (natural light or facing a window), clean mouth, lip products removed if possible, mirror nearby if helpful, and a minute or two set aside. These steps mirror in-app copy and improve focus and framing.

2. **Photo capture (up to five views)** — **Front view** is required. **Right cheek, left cheek, tongue, and palate** can be captured or skipped (optional views have a skip control). The app uses the device camera; each image gets an on-device **quality** estimate. Very poor quality or likely non-oral images may be excluded or down-weighted later.

3. **Review** — The user confirms the set of photos (or retakes specific views if needed).

4. **Analyze** — Starts on-device inference. Processing continues while the next step runs.

5. **Tobacco / exposure questionnaire (ERI)** — Short questions about tobacco use, answered while analysis runs where possible. Answers can adjust **recommendation text** (exposure-informed messaging); they do not replace the core binary screening rule.
6. **Results** — **Risk level** (Low / Moderate / High), **urgency**, **recommendations**, and (when shown) **Screening Risk Score**, **display label**, and explanation text. The user can open **detailed / per-photo** information where the product provides it.

Clinician tip: For validation or audit, note the **time order** you prefer (exam first vs. images first) in your protocol; the app does not enforce clinical examination, but your workflow should make clear whether images are independent of your verbal impression.

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3. What the user sees (single photo)

For each photo, the app shows:

Item	Meaning
Risk level	**Low**, **Moderate**, or **High** — from binary logic (and session logic for multi-photo).
Display label	The 5-class label (e.g. OPMD, OCA, Benign) or “Uncertain” if the model is not confident enough.
Screening Risk Score	P(abnormal) from the binary check (e.g. 85.7%).
Summary / label	5-class label when aligned; when binary and 5-class disagree strongly, only **Retake or further analysis needed** may be shown (no visual-detail line).
Verdict explanation	Short plain-language explanation of the result and suggested next steps.

If the two checks **disagree** in the problematic way (binary positive, 5-class Healthy/Benign), the app shows the risk badge, Screening Risk Score, and **Retake or further analysis needed** only —by design, to avoid implying a benign subtype when screening is positive.

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4. How risk and urgency are decided (current screening policy)

Screening decision is driven **only by the binary model** (the 5-class model does not override it).

Per photo

- Each **usable** photo has a **Screening Risk Score** = binary P(abnormal).

- Per-photo **risk badge** uses binary thresholds: for example **≥ 0.65** → High; **≥ 0.4735 and < 0.65** → Moderate; **< 0.4735** → Low (exact wording in the app may vary slightly by build).

Session (single or multi-photo)

- **Hybrid rule:** Over all usable photos, **maxScore** = highest binary score, **secondScore** = second highest, **hybridAvg** = (maxScore + secondScore) / 2.
- **Screen positive** if **maxScore ≥ 0.65** *or* **hybridAvg ≥ 0.4735**; otherwise screen negative (then retake band / quality handling may suggest retake or low risk as appropriate).
- **Overall risk** combines **screening outcome** with **subtype** from the 5-class path when available: e.g. review + OCA-dominant → High; review + OPMD or benign (or none) → Moderate.
- **Urgency:** Low risk → Routine (about 2–4 weeks); Retake → Retake photos; Review + OCA → Urgent (within about 1 week); Review + other → Prompt (about 1–2 weeks).

The **5-class model** chooses **subtype** (OCA/OPMD/benign) and **label confidence** only. When screen-positive but the aggregated 5-class label is Healthy or Benign, subtype and label confidence are suppressed so patients are not shown a benign subtype alongside a positive screen.

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5. Multiple photos (two or more)

When the user submits multiple photos (front, cheeks, tongue, palate):

1. **Usable photos** — Oral, analyzed, **quality ≥ 0.5** (sub-threshold images may be treated as unclear or excluded from aggregation). Images that look like **strong "Unknown" / non-oral** (high unknown-class probability) can be excluded.

2. **Hybrid rule** — **maxScore** and **secondScore** as above; **hybridAvg** = (maxScore + secondScore) / 2. **Screen positive** if maxScore ≥ 0.65 or hybridAvg ≥ 0.4735.

3. **Label confidence and subtype** — When screen positive, 5-class probabilities are aggregated over **abnormal photos only** (binary ≥ 0.4735), with weights from quality and confidence. That yields label confidence (e.g. "50% (OPMD)") and subtype. When screen negative, aggregation uses all usable photos.

4. **Session-level suppression** — If screen positive but aggregated 5-class is Healthy or Benign, the app does **not** show the "lesion subtype" style detail or the label confidence stat.

Net result: one **overall risk**, **urgency**, **probabilities** when shown, and **recommendations**.

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6. Recommendations and urgency

Recommendations follow the **final** risk level after the policy above and **ERI** (tobacco exposure) when present:

- **Low risk** — No concerning signs detected; routine oral hygiene and dental check-up; rescreen in about 3 months (as shown in app).
- **Moderate risk** — Changes need attention; see dentist within about 2–4 weeks; avoid tobacco/alcohol; monitor.
- **High risk** — Concerning changes; urgent evaluation by oral surgeon or oncologist within about 1 week; bring the report.

Urgency is determined from outcome + subtype (Section 4). The app may add notes (e.g. highlighting **Front view**) when certain views contributed to Moderate or High risk.

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7. Evidence, limitations, and why binary screening leads

Patient-wise evaluation (why older "accuracy" figures can mislead):
Models were re-trained and tested with **patient-level splits**: images from the **same person** do not appear in both training and test. Older **image-wise** splits allowed the same patient in train and test, so metrics could be **inflated** (lighting, anatomy, camera habits). **Patient-wise** metrics are a more honest guide to **new patients**.

What that means in practice:

- **Binary screening** is the main, calibrated lever for "refer vs not" at fixed thresholds. The **0.4735** hybrid threshold is chosen as an **~80% specificity** operating point on an internal pooled frozen test set (**TM_v3**); **0.65** catches a single very confident positive view. These values are **configuration**, not universal clinical truth—they should be reviewed when models or datasets change.
- **5-class** detail is **more sensitive to domain shift** (different cameras, sites, or labeling styles). The app uses it for **interpretability and urgency wording**, not to overturn binary screening.
- Some builds use an **SSL-enhanced** backbone (extra oral-related pretraining); deployment still relies on **calibrated thresholds**, not a raw 0.5 cutoff alone.

The app is a **screening adjunct**, not a substitute for clinical examination, histopathology, or specialist diagnosis.

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8. Thresholds and quality rules (reference)

All of the following live in app configuration (`ScreeningConfig` in the codebase) and may change with a release:

Topic	Typical values	Role
Hybrid screen-positive	hybridAvg ≥ **0.4735**	Session-level positive if max of top two averages high enough.
Single strong positive	maxScore ≥ **0.65**	Session positive regardless of second photo.
Per-photo “abnormal” for aggregation	binary ≥ **0.4735**	Used when weighting 5-class inputs for positives.
Retake suggestion band (per-photo context)	about **0.45–0.50**	Narrow band where the app may suggest clearer photos (exact behaviour depends on session state).
Minimum quality for “usable”	**0.5**	Below this, photos may be excluded or treated as unclear.
Strong “unknown / non-oral”	unknown probability > **0.6**	Can drop a photo from oral screening aggregation.
Unclear image quality	quality < **0.6**	May be flagged as unclear at capture.
Architecture	MobileNetV3-Large–class backbones	Binary + 5-class **ensembles** as shipped in the build.

9. How clinicians can use the app (including tobacco-exposed patients)

1. **Complete the oral exam you would normally perform** first or in parallel, per your protocol—then capture images that **represent** the areas you assessed (buccal mucosa, tongue, floor of mouth, palate, etc.). The app is only as good as the **views and quality** of the photos.
2. **Follow in-app guidance** on lighting, focus, and number of views. Poorly lit or out-of-frame images may be excluded or down-weighted.
3. **Treat binary screen-positive / risk level as a screening signal**, not a diagnosis. Discordance (app positive, you low concern—or the reverse) is **expected sometimes** and is what validation studies measure.
4. **Use 5-class labels cautiously** for patient communication; they are **subordinate** to binary screening and can disagree when anatomy or image domain differs from training data.
5. **Document tobacco exposure** in the app if your study uses it (ERI), so recommendations stay aligned with the intended workflow.
6. **For retake or uncertain states**, follow your clinical safety net: **clinical findings override the app** when they conflict.

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10. Validation against clinician judgment (hospital / controlled studies)

If your institution runs a **prospective validation**—clinicians who routinely examine **tobacco-using** patients also use the app and you analyze **agreement** with clinical decisions—consider:

Aim	Practical approach
Compare screening disposition	Record your **binary** decision (e.g. reassure vs refer / biopsy / specialist) on the **same visit** as the app session; compare to app screen-positive vs screen-negative.
Compare risk gradation	Optionally map your impression to Low / Moderate / High concern and compare to app **risk level** (scales are not identical).
Handle disagreement	Predefine that **clinical judgment** governs patient management; the app is under evaluation. Capture disagreements for analysis, not as isolated “errors.”
Reduce bias	Decide whether image capture is **independent** of your verbal result (e.g. assistant captures before you state conclusion) if the study requires it.
Correlation metrics	Agreement (kappa), sensitivity/specificity of app vs clinician **as one reference** (neither is a gold standard without pathology), calibration of **Screening Risk Score** vs your ordinal concern.
Case mix	Record exposures (tobacco type, pack-years if used), sites photographed, and reason for visit.

Framing validation around **binary agreement** under **patient-wise** model behaviour matches how the product is designed: binary screening is the robust layer; 5-class is auxiliary for wording and subtype.

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11. Summary for clinicians

Question	Answer
What is the main “result”?	**Risk level** (Low / Moderate / High), **urgency**, and **recommendations**, plus the 5-class **label** when shown and explicit handling when models disagree.
How is risk computed?	**Binary** model drives screening (hybrid: maxScore ≥ 0.65 or hybridAvg ≥ 0.4735 → screen positive). **Overall risk** combines outcome + subtype (e.g. review + OCA → High; review + OPMD/benign → Moderate).
Why two checks?	Binary is tuned for refer vs not; 5-class gives interpretable labels and subtype for wording/urgency. The binary **is not** overridden by the 5-class.
What if the two checks disagree?	The app shows the **risk badge** and **Screening Risk Score**, and **“Retake or further analysis needed”** without implying a benign subtype.
Is the AI diagnostic?	**No**—screening adjunct; clinical examination and standard pathways remain authoritative.
Why trust patient-wise development?	Train/test splits avoid **same-patient leakage**, so reported performance is closer to **new-patient** behaviour than older image-wise figures.
Do thresholds ever change?	Yes—thresholds are **versioned configuration**. After model or app updates, confirm behaviour against **release notes** or your institution’s validation SOP.